

Hello

I'm Ammar Elmesaly

Mansoura, Egypt | +20 10 6313 5372 | ammar-elmesaly@proton.me

Portfolio elmesaly.dev | GitHub: github.com/ammamr-elmesaly | LinkedIn: linkedin.com/in/ammamr-elmesaly

Skills

- **Languages:** TypeScript, JavaScript (ESM), Rust, SQL
- **Backend & Databases:** NestJS, Node.js, Express.js, PostgreSQL, TypeORM, MongoDB, REST APIs
- **Frontend & Desktop:** React, Electron, Vite
- **Tools & Environment:** Linux (Fedora), Git, GitHub Actions
- **Spoken Languages:** Arabic (Native), English (Fluent)

Technical Projects

12 Aug, 2026 - Present

[Personal Portfolio \(elmesaly.dev\)](https://elmesaly.dev)

- Built a developer portfolio and blog using Next.js (App Router), TypeScript, Tailwind CSS, and MDX
- Designed fluid, declarative UI animations and layout transitions using Framer Motion to enhance user experience
- Implemented a dynamic MDX content pipeline for technical articles, featuring custom metadata parsing and static site generation (SSG)
- Configured automated serverless deployment to Vercel, optimizing asset delivery and edge performance

28 Jun, 2026 - 25 Jul, 2026

[Installments System](#)

- Built a modular **NestJS** backend to generate and track customer installment payment plans
- Designed **PostgreSQL** schemas using **TypeORM**, leveraging **ACID transactions** to prevent payment race conditions
- Managed database updates across environments using version-controlled **migrations**
- Developed a midnight **Cron job** to identify overdue payments and alert admins via **Telegram Bot API**
- Implemented multi-tier **RBAC** (Superadmin, Auditor, Collector) and JWT authentication to secure endpoints

27 Dec, 2025 - 31 Mar, 2026

[DVLD App](#)

- Designed a comprehensive **TypeScript monorepo** featuring shared logic across Client, Server, and **Electron** environments
- Designed relational PostgreSQL schemas to track driver history, license statuses, and application pipelines
- Developed a specialized UI for administrative workflows

14 Sep, 2025 - 28 Oct, 2025

[Quran Api](#)

- Designed and deployed a high-performance REST API using **Node.js** and **Express** with **TypeScript**
- Integrated **PostgreSQL** to serve structured religious text data with high availability
- Focused on clean code principles and **API documentation**, ensuring the API is easily consumable for third-party developers

25 Sep, 2025 - 1 Oct, 2025

Rust TUI Todo Manager

- Built a high-performance **Rust TUI** using **Crossterm** for low-latency, keyboard-driven task management
- Engineered a custom data persistence layer for efficient terminal-based **CRUD** operations
- Leveraged Rust's **ownership model** to ensure a memory-safe and crash-free architecture

Education

October 2024 - July 2028

Mansoura University, Mansoura, Egypt – *Computer Science*